

MARIA PAUL THURKDAYIL

DATA SCIENCE STUDENT



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SUMMARY

Master's student in Data Science at Chennai Mathematical Institute with a strong foundation in Computer Science and Engineering. Experienced in developing LLM-powered pipelines. Proven ability to bridge research and production through open-source contributions and end-to-end model deployment. Seeking Data Science and Machine Learning roles.

EDUCATION

Chennai Mathematical Institute, Masters in Data Science

Aug 2024- May 2026

EXPIERENCE

Research Intern | IIT Jammu) | Jun 2026 – Present

<https://iitjammu.ac.in/mira/>

- **Synthia (Synthetic Data Generation Pipeline):** Optimised a Python pipeline to generate instruction fine-tuning data (context, instruction, response triples) from raw text documents using self-hosted LLMs.
- Configured and validated the local development environment, establishing secure connectivity to MIRA's private GPU inference cluster via Tailscale VPN.
- **Gaperon (Multilingual Text Quality Classifier):** Optimised a classification system using jhu-clsp/mmBERT-base with 6 independent heads to evaluate text quality (clarity, coherence, depth, grammar, usefulness, overall) across English, Hindi, and Tamil.
- Built an end-to-end workflow using SkyPilot and HuggingFace Hub for dataset streaming, checkpoint versioning, and seamless model promotion across branches.

Research Intern, S3 Labs, IIT Bhilai | May 2025- July 2025

- **Technical Image Understanding:** Evaluated LLM inference capabilities on Packet, Sequence, and State Diagrams by converting Mermaid Code Diagrams and hand-drawn images into structured representations.
- **Legal Infographic Generation:** Prototyped an LLM-powered application using LangChain to automate the extraction and visualization of key arguments and timelines from complex legal documents.
- Conducted comprehensive literature reviews on LLM applicability in the legal domain, utilizing the IL-TUR dataset (IIT Kanpur) for experimentation.

PROJECTS

Dyanamic Reflections: Probing Video Representations with Text Alignment | June 2026

- Partnered with a teammate to evaluate and present cutting-edge research on multimodal video-text representation alignment and spatio-temporal semantics for the "Introduction to Generative Intelligence" course at CMI.
- Explored accompanying Jupyter Notebook implementations of zero-shot probing techniques and analyzed cross-modal semantic alignment methodologies, culminating in a comprehensive technical presentation of the findings.

LLM Distillation for Named Entity Recognition, Applied Machine Learning Project | May 2026

- Collaborated with a teammate to develop a knowledge distillation pipeline, compressing a 12B-parameter LLM (Gemma 3) into a 270M-parameter model for fine-grained NER.
- Helped validate the final model, which achieved a strict-match F1 score of 0.698, successfully matching the teacher model's performance at a fraction of the compute cost.

Differential Privacy in PyTorch (Opacus) | Data Quality and Data Privacy Project | May 2026

- Investigated and resolved a critical privacy bug involving silent epsilon miscalculations when using WeightedRandomSampler in Meta's open-source differentially private deep learning library.

Perspective-Aware Sentiment Analysis | Applied Data Analytics Project | Nov 2025

- Built a real-time data pipeline using Google News RSS and feedparser to fetch headlines based on dynamic search queries and time windows.
- Engineered a perspective-aware sentiment engine using GPT-4o-mini that adapts analysis based on user-defined personas, improving upon a baseline twitter-roberta-base-sentiment-latest model.

Predicting Crime Categories | XGBoost, LightGBM | Jun 2024 – Aug 2024

- Developed a VotingClassifier ensemble model to predict 6 crime categories from a dataset of 20,000+ records featuring location, demographic, and temporal variables.
- Achieved an accuracy of 88.08%, significantly outperforming the baseline accuracy of 58%.

Forecasting Marine Fish Production in Kerala , Random Forest, MLR, MLP | Jan 2022 – May 2022

- Analyzed 54 years of marine fish production data (CMFRI) and 9 environmental variables (ICOADS) to build predictive models.
- Optimized a Random Forest model achieving an R^2 score of 0.87, MAE of 0.05, and RMSE of 0.07, demonstrating the highest predictive power among tested algorithms.

Chat Application- Publisher Subscriber Architecture | November 2019

- Developed a real-time communication system using the MQTT protocol, applying a publisher-subscriber model to support chat-style applications as part of Hack-Kochi#2

TOOLS & LIBRARIES

Data Science Libraries: TensorFlow, OpenCV, scikit-learn, NumPy, pandas, matplotlib, seaborn

NLP & Gen- AI : Hugging Face Transformers, LangChain, LLM Fine-tuning, Prompt Engineering

Statistical Tools: R, SPSS, statsmodels

Data Analytics: Python, Power BI, MS-Excel

RDBMS: MySQL, PostgreSQL, MongoDB, Redis

Big Data Frameworks: Apache Spark, Hadoop

Model Deployment Frameworks: Flask, Streamlit , Skypilot, Docker , Tailscale

TEACHING & MENTORING

Teaching Assistant, NPTEL

Jan 2024- April 2024

noc24-cs47: The Joy of Computing Using Python.

Instructor: Dr Sudarshan Iyengar, Head of Computer Science Department, IIT Ropar

Prepared weekly assignments & handled the forum for the course having **60,000** registered learners.

Student Mentor, IITM BS Degree Programme

Mar 2024- April 2024

BSMA1004: Statistics for Data Science 2

Instructor: Andrew Thangaraj, Professor, Electrical Engineering, IIT Madras

Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of **25 students**.

OTHERS

1. Winner Hack Kochi #2

November 2019

24-hour Hackathon for girls organised by IEEE Kochi Hub.

2. KEAM Rank 333

August 2018

Kerala State Engineering Entrance Exam.

VOLUNTEERING

Covid-19 Response Volunteer, Caregiver Saathi

May 2021

ONLINE CERTIFICATIONS

- Google IT Support Specialization, Coursera
- Introduction to Linux, The Linux Foundation
- Machine Learning Foundations: A Case Study Approach
- SQL For Data Science